DR. KATHERINE HAWTHORNE

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PROFESSIONAL SUMMARY

Distinguished Software Architect and Research Fellow with 15+ years of foundational contributions to distributed consensus protocols, high-availability cloud platforms, and large-scale enterprise system transformations. Frequent keynote speaker and author of 6 peer-reviewed papers on distributed systems reliability.

PROFESSIONAL EXPERIENCE

Staff Platform Engineer — Apex Cloud Solutions

2021-03-01 – Present

- Spearheaded architectural re-platforming of legacy monolithic order processing system into 14 distributed Go microservices handling 45,000 req/sec at peak.
- Reduced P99 API latency from 420ms to 28ms by introducing Redis cluster caching layer and gRPC transport protocols.
- Architected automated Kubernetes deployment manifests and Helm charts deployed across 3 AWS multi-region clusters.
- Mentored 8 junior and mid-level engineers, running weekly architecture review sessions and code craftsmanship workshops.

Senior Software Engineer — Catalyst Financial Technologies

2018-06-01 – 2021-02-28

- Built and maintained PCI-compliant credit card processing pipeline managing $850M in annual transactional volume.
- Implemented zero-downtime database schema migrations for PostgreSQL tables containing over 250 million records.
- Developed end-to-end audit logging infrastructure using Apache Kafka and Elasticsearch for regulatory compliance.

Full Stack Developer — Kinetic Web Labs

2017-06-01 – 2018-05-31

- Engineered customer-facing React SPA dashboard integrated with Express and Node.js REST API endpoints.
- Optimized Webpack bundle sizes by 44% through code-splitting, tree-shaking, and lazy-loading of secondary routes.

Principal Research Scientist — Global Defense & Aerospace Systems

2014-01-01 – 2017-05-30

- Directed research team investigating Byzantine fault-tolerant consensus mechanisms for satellite mesh networks.
- Formally verified safety invariants of state machine replication algorithms using TLA+ and formal model checking.
- Awarded 2 US patents on secure multi-hop routing protocols in high-interference electromagnetic environments.

Senior Systems Engineer — Siemens Industrial Automation

2010-09-01 – 2013-12-31

- Designed real-time deterministic Ethernet drivers for programmable logic controllers (PLCs).
- Achieved sub-microsecond clock synchronization accuracy adhering to IEEE 1588 Precision Time Protocol.

Member of Technical Staff — Bell Laboratories

2008-06-01 – 2010-08-31

- Researched optical switching architectures for core telecommunications backbones.
- Co-authored 4 technical whitepapers on dynamic wavelength allocation and network topology optimization.

EDUCATION

University of Illinois Urbana-Champaign, M.S. in Computer Science 2015-08-15 – 2017-05-20

GPA: 3.91

Purdue University, B.S. in Computer Engineering 2011-08-20 – 2015-05-10

GPA: 3.78

KEY PROJECTS

Distributed Task Queue Engine | *Go, Redis, Docker, Prometheus* 2023-01-01 – 2023-08-30 github.com/marcusvance/task-queue

- High-throughput async job worker system with Redis persistence, backpressure handling, and exponential backoff retry semantics.

Event Horizon Analytics | *Node.js, PostgreSQL, WebSockets* 2022-03-01 – 2022-11-15 github.com/marcusvance/event-horizon

- Real-time telemetry aggregation engine processing 100k events/sec with WebSockets and TimescaleDB analytics.

Raft-KV Distributed Store | *Rust, Tokio, gRPC* github.com/hawthorne/raft-kv

- Production-ready distributed key-value store implementing Raft consensus with log compaction and dynamic membership changes.

TLA+ Model Repository | *TLA+, PlusCal* github.com/hawthorne/formal-specs

- Curated collection of formal specifications for modern cloud distributed consensus and two-phase commit variants.

TECHNICAL SKILLS

Languages: TypeScript, Go, Python, SQL

Frontend: React, Next.js, Tailwind CSS

Backend: Node.js, PostgreSQL, Redis

DevOps & Cloud: Docker, Kubernetes, AWS, Terraform

HONORS, AWARDS & CERTIFICATIONS

- **AWS Certified Solutions Architect – Professional** — Amazon Web Services (2023)
- **Certified Kubernetes Administrator (CKA)** — Linux Foundation (2022)
- **ACM Distinguished Engineer Award** (2023): Awarded for pioneering contributions to distributed system resilience
- **Best Technical Paper – IEEE CloudCom** (2020): Selected among 350 submissions for work on sub-millisecond failover protocols